

BAUM TenPers Bitcoin Hint

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Interpreting the β -bound and its link to partial correlation

Bound. Let e_{t+1} denote the *idiosyncratic wedge* in stocks relative to crypto (after controls), and let ε_{t+1}^k be the *idiosyncratic crypto* component. The OLS slope of (post-control) stocks on crypto satisfies

$$1 - \sqrt{\frac{\text{Var}(e_{t+1})}{\text{Var}(\varepsilon_{t+1}^k)}} \leq \beta \leq 1 + \sqrt{\frac{\text{Var}(e_{t+1})}{\text{Var}(\varepsilon_{t+1}^k)}}.$$

Write $\rho := \text{Var}(e_{t+1})/\text{Var}(\varepsilon_{t+1}^k)$. Then $\beta \in [1 - \sqrt{\rho}, 1 + \sqrt{\rho}]$, so the *maximum distance* of β from 1 is exactly $\sqrt{\rho}$.

Meaning.

- If ρ is small (wedge variance tiny relative to crypto's idiosyncratic variance), β is *forced* to be near 1: post-control stock and crypto move almost one-for-one.
- If ρ is large, β is *allowed* to wander far from 1: co-movement can weaken materially.

When the bound is tight. Using $\beta = 1 + \frac{\text{Cov}(e_{t+1}, \varepsilon_{t+1}^k)}{\text{Var}(\varepsilon_{t+1}^k)}$ and $|\text{Cov}(e, \varepsilon^k)| \leq \sqrt{\text{Var}(e) \text{Var}(\varepsilon^k)}$ (Cauchy-Schwarz), the endpoints are attained when e_{t+1} is perfectly proportional to ε_{t+1}^k :

$$\beta = 1 + \sqrt{\rho} \quad \text{if } e_{t+1} = +c\varepsilon_{t+1}^k, \quad \beta = 1 - \sqrt{\rho} \quad \text{if } e_{t+1} = -c\varepsilon_{t+1}^k$$

for some $c > 0$. If $\text{Cov}(e_{t+1}, \varepsilon_{t+1}^k) = 0$, then $\beta = 1$ (the center of the interval).

Link to the partial correlation r . Let $\sigma_E^2 := \text{Var}(\varepsilon_{t+1}^k)$ and $\sigma_e^2 := \text{Var}(e_{t+1})$. For residuals (post-control series),

$$r = \frac{\beta \sigma_E}{\sqrt{\beta^2 \sigma_E^2 + \sigma_e^2}}$$

so two forces push r *down*: (i) a larger wedge variance σ_e^2 (inflates the denominator), and (ii) a β that sits far from 1 (which only happens when ρ is large). Thus the same variance ratio ρ that widens the admissible set for β also depresses r .

Numeric intuition (set $\sigma_E = 1$).

$$\text{Small wedge: } \rho = 0.1 \Rightarrow \beta \in [0.684, 1.316], \quad \beta = 1, \quad \sigma_e^2 = 0.1 \Rightarrow r = \frac{1}{\sqrt{1+0.1}} \approx 0.953.$$

$$\text{Large wedge: } \rho = 2 \Rightarrow \beta \in [-0.414, 2.414], \quad \beta = 1, \quad \sigma_e^2 = 2 \Rightarrow r = \frac{1}{\sqrt{1+2}} \approx 0.577.$$

Same β but larger wedge $\Rightarrow$ much smaller r .

Implication for the “4 up / 4 down” result. Across countries, post-2019 changes in the variance ratio ρ can differ: if ρ *falls* (wedge shrinks), the β -interval tightens around 1 and r rises (BTC looks more equity-like); if ρ *stays high* or increases, the interval stays wide and r remains low or falls (BTC appears more hedge-like). Hence opposite movements in r across countries are fully consistent with the model.